

Alex Clunan

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EDUCATION

University of Illinois Urbana Champaign

Urbana-Champaign, IL | Expected Graduation: May 2028

- *Master of Science in Electrical Engineering*

University of Virginia School of Engineering & Applied Science

Charlottesville, VA | 2022 - 2026

- *Bachelor of Science in Electrical Engineering with Highest Distinction | GPA: 3.94*

RESEARCH EXPERIENCE

Biophysical Systems Lab - University of Virginia

Charlottesville, VA | Jan 2025 - Present

- **Real-Time Biological Cell Sorting:** Built custom instrumentation for near-sensor impedance signal processing that isolates rare cells from liquid biopsies in real time, enabling researchers to study their role in cancer, neurodegeneration, and metabolic disease and to develop targeted therapeutics.

Custom Multi-Frequency Impedance Analyzer (Zynq-7000 SoC)

Jan 2026 - Present

- Architected and delivered a 4-channel multi-frequency impedance analyzer with an integrated real-time cell-sorting trigger on a Xilinx Zynq-7020 SoC and custom 10-layer mixed-signal PCB, meeting a <5 ms detection-to-actuation latency requirement; two-person capstone continued as funded research. Publication set for mid-2027.
- Owned the complete FPGA design: custom Verilog/SystemVerilog RTL integrated with Xilinx IP - modules include DDS excitation, LVDS SerDes interfaces to a 16-bit/200 MSPS ADC and 16-bit DAC, quadrature demodulation (digital lock-in amplifier), CIC + reloadable-coefficient FIR decimation chain, CORDIC magnitude/phase extraction, AXI4-Stream resizable packetizer feeding AXI DMA, and a 16×32-bit AXI4-Lite register file.
- Closed timing across two clock domains (200 MHz / 100 MHz) using AXI-Stream CDC FIFOs, RTL level pipelining, BRAM utilization, and synchronized per-domain reset generation; debugged with Vivado ILA and xsim.
- Developed embedded Linux firmware in C under PetaLinux on the dual-core Cortex-A9 PS: UDP/Ethernet data, control, and trigger protocols; DMA kernel-module interface; SPI configuration and bring-up of both data converters; rolling-baseline peak detection with phase-window event classification.
- Built a Python based GUI including a live scope display, register-level hardware introspection, and a filter designer
- Validated end-to-end on a live impedance-cytometry setup, cross-checking against a commercial impedance analyzer to resolve single-cell impedance signatures;

Raspberry Pi Cell Triggering System

Jan 2025 - Aug 2026

- Built the predecessor sorting system on a Raspberry Pi with a commercial impedance analyzer, processing cell impedance streams in Python to actuate a sorting valve at <70 ms latency with <1 ms timing variation; verified timing with an oscilloscope, arbitrary waveform generator, and live cell microfluidic experiments.

iSENS Lab - University of Illinois Urbana Champaign

Urbana-Champaign, IL | Aug 2026 - Present

- NASA-funded canopy-penetrating FMCW radar for forestry sensing: developing the FPGA signal chain on a 4x2 RFSoc (direct-RF sampling, range-Doppler FFT pipeline) and specifying analog front-end components;

JOB EXPERIENCE

University of Illinois Urbana Champaign

Aug 2026 - Present

Teaching Assistant (TA) - ECE 110 Introduction to Electronics

- Serving as a Lab TA in two 3 hour sections, lecturing students, grading assignments, and overseeing lab activities.

University of Virginia Electrical and Computer Engineering Department

Aug 2023 - May 2026

Teaching Assistant (TA) - Applied Physics: Electricity and Magnetism, FPGA Digital Design, Computer Architecture and Design

- Teaching Assistant: Graded assignments, held office hours, ran labs, created answer keys, and coordinated with the instructor. Mentored topics include FPGA design, Verilog, multistage CPU design, verification, and E&M physics.
- Promoted to Head TA (Physics) (8/2024) - Managed a team of 15 TAs, standardized grading turnaround by reducing feedback time from a maximum of 4 weeks to 7 days across assignments, and designed homework and lab exercises. Created a helpdesk email system that centralized requests for 480+ students.

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Ultrata LLC

Remote | Jun 2025 - Aug 2025

Software Engineering Intern

- Benchmarked performance data of Ultrata's S3 interface vs. MinIO using a proprietary benchmarking tool.
- Modified the C standard library for compatibility with the Intelligent Memory Fabric (IMF), modifying key library functions.
- Created a C++ allocator class to enable compatibility of C++ STL Vectors with IMF.
- Developed a fully functional Linux FUSE filesystem from scratch that interfaces with IMF in C/C++.

PROJECTS

8-Bit 2-GSPS 45nm Time-Interleaved SAR ADC

Spring 2026

- Co-designed and verified a $9\times$ time-interleaved SAR ADC in Cadence at 2 GS/s in 45nm CMOS, achieving 5.58 ENOB and 42.2 dBc SFDR near Nyquist across a full FFT-based characterization.
- Built each sub-channel from a bootstrapped sample-and-hold, PMOS-input StrongARM comparator, and 4+4 segmented CDAC; segmentation cut total array capacitance $\sim 32\times$ versus a binary-weighted 8-bit array, reducing settling time and switching energy.

Scam Shield - HooHacks 2026

March 2026

- Won 3rd place out of 220+ teams at HooHacks 2026 building ScamShield, a real-time AI scam detection device targeting the elder fraud market.
- Built the full hardware and AI pipeline - integrating live audio processing, AI scam scoring, and multi-channel family alerts using a Raspberry Pi and connectable hardware modules within a 24-hour hackathon timeline.

Spiking Neural Network (SNN) Performance Analysis on the Akida PCIe Board

Fall 2025

- Co-created and trained a CNN on MNIST and converted it to an SNN to compare performance on neuromorphic hardware.

High-Speed 64kb Cache SRAM, FreePDK 45nm

Spring 2025

- Designed and verified a 64kb 32-bit-word cache SRAM achieving a 6.40×10^{-31} figure of merit (energy $\times$ delay² $\times$ area $\times$ idle power) - the best result of any project reviewed from prior offerings of the course
- Optimized 6T bitcell sizing, sense amplifier selection, decoder topology, and array aspect ratio through parametric SPICE sweeps; StrongARM sense amp chosen over a 13T latch-type design for a 28x read-speed improvement (24 ps vs. 668 ps)
- Implemented selective wordline overvolting on the two farthest columns to cut worst-case access delay without the power cost of overvolting the full array
- Verified functionality at 625 MHz across all process corners, $VDD \pm 0.1$ V, and 0-50 °C

RISC-V CPU

Spring 2025

- Developed a combined RISC-V CPU and UART Transceiver with support for arithmetic, logical, memory, branch, and jump instructions. Verified functionality with HDL testbenches and an assembly program using ModelSim.

Field Programmable Gate Array (FPGA) Design Projects

2025

- Implemented PONG in Verilog from scratch using pushbutton controls and a VGA output as a personal project.
- Created a two way AES-128 text based two-way encrypted radio communication system using multiple PYNQ-Z1 boards.
- Designed an FPGA based synthesizer with an LFO, mixer, different waveform settings, and a PWM DAC output.

Skills - C, C++, SystemVerilog, Verilog, Python, Quartus, Vivado, Vitis, Cadence Virtuoso, Linux, gdb, Git, MATLAB, ModelSim, RF DSP, Altium, I2C, SPI, LVDS, Jira

Courses - Data Converters, Computer Architecture & Design, AI Hardware Software Co-Design, Digital Signal Processing, AI Hardware, FPGA Design Lab, Hardware for Modern Day Computing, Introduction to VLSI Design

Certifications - Student Pilot Airplane Single Engine Land, Virginia Emergency Medical Technician (EMT).